

Auxiliary Material for “Towards Causal Item–Item Link Mining in Organic CF Data”

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This document provides auxiliary material for the submission “Towards Causal Item–Item Link Mining in Organic CF Data”. It includes additional details for the ATE standard-deviation estimator (Sec. 1), full synthetic-data generation and interventional-ground-truth details (Sec. 2), the complete LLM labeling protocol and summary statistics (Sec. 3), the full specification of the outcome-model (OM) baseline (Sec. 4), and additional results omitted from the main paper for space (Sec. 5). Section and equation references to the main paper are used where relevant.

Our code is available at: <https://anonymous.4open.science/r/Causall2I-8CC4/README.md>.

1 Additional Details for the ATE Standard Deviation Estimator

This section provides additional details for the standard-deviation estimator used in Eq. (8) of the main paper for the unstabilized IPW ATE in Eq. (7). In the current main-paper setting, the unstabilized estimator is a simple difference of sample means, so its uncertainty can be computed directly from the variance of a weighted per-user contrast. We then extend the discussion to stabilized IPW, where the ratio-of-means form requires a more careful delta-method treatment.

Interpretation (ATE/STD). The ratio $\widehat{ATE}_{i \rightarrow j} / \widehat{STD}_{i \rightarrow j}$ (our $\widehat{STE}_{i \rightarrow j}$) can be interpreted as an uncertainty-normalized effect size: it measures how many estimated standard-deviation units the effect lies above the null hypothesis of *no directed causal influence* ($ATE_{i \rightarrow j} = 0$). Equivalently, it is a discovery-oriented signal-to-noise score: large values indicate effects that are not only high-magnitude in probability units (ATE), but also supported by sufficient overlap and effective sample size. This normalization is particularly important in sparse organic logs, where IPW estimates can be dominated by a small number of users with extreme inverse-propensity weights; dividing by $\widehat{STD}_{i \rightarrow j}$ down-weights such high-variance links and prioritizes item pairs whose estimated influence is both large and precisely estimated.

1.1 Unstabilized IPW: direct standard-deviation estimator

Fix an ordered pair (i, j) and consider the N evaluation users $u = 1, \dots, N$. Let $T_u \in \{0, 1\}$ indicate whether user u consumed item i in the post-cutoff window (treatment), and let $Y_u \in \{0, 1\}$ indicate whether u subsequently consumed item j in that window (outcome), as defined in Sec. 3 of the main paper. Let $\pi_u \in (0, 1)$ be the estimated propensity $\Pr(T_u=1 \mid Z_u)$.

Define the per-user treated and control contributions

$$A_u = \frac{T_u Y_u}{\pi_u}, \quad C_u = \frac{(1 - T_u) Y_u}{1 - \pi_u}.$$

Then the unstabilized IPW estimator from Eq. (7) of the main paper can be written as

$$\widehat{\tau} = \frac{1}{N} \sum_{u=1}^N A_u - \frac{1}{N} \sum_{u=1}^N C_u. \quad (1)$$

Equivalently, if we define the weighted per-user contrast

$$Z_u \triangleq A_u - C_u = \frac{T_u Y_u}{\pi_u} - \frac{(1 - T_u) Y_u}{1 - \pi_u}, \quad (2)$$

then

$$\widehat{\tau} = \frac{1}{N} \sum_{u=1}^N Z_u. \quad (3)$$

Thus, unlike stabilized IPW, the unstabilized estimator is simply the sample mean of the weighted contrasts Z_u .

Its variance therefore takes the standard sample-mean form

$$\text{Var}(\widehat{\tau}) = \frac{1}{N} \text{Var}(Z_u). \quad (4)$$

A direct plug-in estimator is obtained by replacing $\text{Var}(Z_u)$ with the sample variance of the observed contrasts:

$$\widehat{\text{Var}}(\widehat{\tau}) = \frac{1}{N(N-1)} \sum_{u=1}^N (Z_u - \bar{Z})^2, \quad \bar{Z} = \widehat{\tau}. \quad (5)$$

We then define

$$\widehat{STD}(\widehat{\tau}) = \sqrt{\widehat{\text{Var}}(\widehat{\tau})}, \quad \widehat{STD}_{i \rightarrow j} \triangleq \widehat{STD}(\widehat{\tau}).$$

For consistency with the notation used in the main paper, Eq. (5) can also be written in covariance form. Let

$$V_u \triangleq (A_u, C_u)^\top, \quad \bar{V} \triangleq \frac{1}{N} \sum_{u=1}^N V_u = (\bar{A}, \bar{C})^\top,$$

and let

$$\widehat{\Sigma} = \frac{1}{N-1} \sum_{u=1}^N (V_u - \bar{V})(V_u - \bar{V})^\top.$$

Then

$$\widehat{\text{Var}}(\widehat{\tau}) = \frac{1}{N} (1, -1) \widehat{\Sigma} (1, -1)^\top, \quad (6)$$

which is exactly the expression used in Eq. (8) of the main paper. Since the unstabilized estimator is a linear contrast, Eqs. (5) and (6) are equivalent.

1.2 Stabilized IPW and the delta-method extension

The current main paper uses the *unstabilized* IPW estimator above. For completeness, we also evaluated a *stabilized* IPW variant, which takes the ratio-of-means form

$$\widehat{\tau}_{\text{stab}} = \frac{\bar{A}}{\bar{B}} - \frac{\bar{C}}{\bar{D}}, \quad (7)$$

with

$$A_u = \frac{T_u Y_u}{\pi_u}, \quad B_u = \frac{T_u}{\pi_u}, \quad C_u = \frac{(1 - T_u) Y_u}{1 - \pi_u}, \quad D_u = \frac{1 - T_u}{1 - \pi_u}.$$

Here the denominators $\bar{B}$ and $\bar{D}$ are themselves random, so $\widehat{\tau}_{\text{stab}}$ is no longer a simple average of per-user contrasts. This is precisely the setting in which the delta method becomes useful.

Define

$$W_u = (A_u, B_u, C_u, D_u)^\top, \quad \bar{W} = \frac{1}{N} \sum_{u=1}^N W_u,$$

and write

$$\widehat{\tau}_{\text{stab}} = g(\bar{W}), \quad g(a, b, c, d) = \frac{a}{b} - \frac{c}{d}.$$

Applying the multivariate delta method [2, 3] gives

$$\text{Var}(\widehat{\tau}_{\text{stab}}) \approx \frac{1}{N} \nabla g(\mu)^\top \Sigma \nabla g(\mu), \quad (8)$$

where $\mu = \mathbb{E}[W_u]$, $\Sigma = \text{Var}(W_u)$, and

$$\nabla g(a, b, c, d) = \left(\frac{1}{b}, -\frac{a}{b^2}, -\frac{1}{d}, \frac{c}{d^2} \right)^\top. \quad (9)$$

Using the empirical covariance

$$\widehat{\Sigma}_W = \frac{1}{N-1} \sum_{u=1}^N (W_u - \bar{W})(W_u - \bar{W})^\top,$$

the corresponding plug-in variance estimator is

$$\widehat{\text{Var}}(\widehat{\tau}_{\text{stab}}) = \frac{1}{N} \nabla g(\bar{W})^\top \widehat{\Sigma}_W \nabla g(\bar{W}). \quad (10)$$

This is the more careful treatment required for stabilized IPW, precisely because of its ratio-of-means structure. In our experiments, however, the stabilized variants underperformed the unstabilized ATE/STE after support filtering and propensity clipping, so the main paper adopts the unstabilized estimator in Eqs. (7)–(8).

1.3 Other auxiliary variants

For completeness, the same four-dimensional delta-method template also applies to the auxiliary ratio-of-means variants, including the propensity-free ablation ABLT and its standardized version ABLT/STD, obtained by substituting the corresponding per-user building blocks into the same construction.

1.4 Practical Notes and Numerical Stability

Positivity/overlap and extreme weights. The unstabilized IPW estimator requires overlap: $0 < \pi_u < 1$ for all users in the evaluation set. In sparse CF logs, some items can yield extreme propensities and high-variance weights. Accordingly, we apply propensity clipping in the main paper when computing Eq. (7).

Computational cost. For the unstabilized estimator, computing Eq. (5) or Eq. (6) is $O(N)$ per pair and adds negligible overhead relative to computing $\widehat{\text{ATE}}_{i \rightarrow j}$ itself. For stabilized IPW and ABLT/STD, the corresponding auxiliary derivations use four-dimensional covariance matrices, which remain computationally trivial in practice.

2 Synthetic Data Generation and Interventional Ground Truth

Here we provide additional implementation details regarding the simulator used in Sec. 6.1 (Simulation Results). The simulator generates implicit user–item interaction logs that combine (i) preference-driven correlations induced by latent user preferences and (ii) a

sparse set of planted directed item–item causal links. Crucially, the simulator supports explicit interventions, enabling oracle ground-truth $\text{ATE}_{i \rightarrow j}^*$ for sanity-checking causal link mining.

2.1 Preference-driven base propensities

Let $\mathcal{U}$ and $\mathcal{I}$ denote the user and item sets, with $|\mathcal{U}|$ users and $|\mathcal{I}|$ items. We first construct a realistic base consumption-probability matrix $S \in [0, 1]^{|\mathcal{U}| \times |\mathcal{I}|}$ that captures preference-driven interaction patterns. In our implementation, we fit a standard matrix factorization (MF) model to a real CF dataset (we used MovieLens-1M [1]), producing user factors $P \in \mathbb{R}^{|\mathcal{U}| \times d}$, item factors $Q \in \mathbb{R}^{|\mathcal{I}| \times d}$, biases b_u, b_i , and global offset μ , and define

$$s_{u,j} = \mu + b_u[u] + b_i[j] + P[u]^\top Q[j], \quad S_{u,j} = \sigma(s_{u,j}),$$

where $\sigma(\cdot)$ is the sigmoid. Sampling interactions from S yields realistic co-consumption patterns driven by latent tastes (i.e., the confounding structure in Fig. 1).

2.2 Planted directed causal links

We encode planted causal influence using a sparse directed matrix $C \in [0, 1]^{|\mathcal{I}| \times |\mathcal{I}|}$. An entry $C_{i,j} > 0$ represents an additional *direct* probability that consuming item i triggers consumption of item j (beyond base preferences). All other entries are zero, so the planted causal graph is sparse. For simplicity, we restrict to nonnegative effects; extensions to signed effects require a different composition rule.

2.3 Generative consumption model

Let A_u denote the set of items consumed so far by user u . We define the probability that user u consumes item j given current history A_u as a noisy-OR composition:

$$p_{u,j} = 1 - (1 - S_{u,j}) \prod_{i \in A_u} (1 - C_{i,j}). \quad (11)$$

Here $S_{u,j}$ captures preference-driven propensity, while each consumed $i \in A_u$ adds an independent “trigger” chance $C_{i,j}$. This form ensures $p_{u,j} \in [0, 1]$ and supports efficient batched computation.

2.4 Phase-based simulation with timestamps

We simulate interactions in discrete phases to mimic cascades of downstream consumption. We maintain (i) a cumulative interaction matrix $W \in \{0, 1\}^{|\mathcal{U}| \times |\mathcal{I}|}$ and (ii) a frontier matrix A containing items newly added in the last phase.

Initialization. We sample an initial set of interactions from base propensities:

$$W_{u,j}^{(0)} \sim \text{Bernoulli}(S_{u,j}), \quad A^{(0)} \leftarrow W^{(0)}.$$

Propagation. At phase $t \geq 1$, only frontier items $A^{(t-1)}$ generate new triggered interactions via C . Define induced probabilities

$$P_{u,j}^{(t)} = 1 - \prod_{i \in A_u^{(t-1)}} (1 - C_{i,j}),$$

sample $Z^{(t)} \sim \text{Bernoulli}(P^{(t)})$, and add only previously unseen consumptions:

$$A^{(t)} \leftarrow Z^{(t)} \odot (1 - W^{(t-1)}), \quad W^{(t)} \leftarrow W^{(t-1)} + A^{(t)}.$$

We iterate until $A^{(t)}$ is empty (no new consumptions) or a maximum phase budget is reached. To produce timestamps, we assign times uniformly at random within each phase, with all events in phase $t+1$ occurring after all events in phase t .

2.5 Efficient batched implementation

We compute the complement-product efficiently in log space. Let $U = \log(1 - C)$ (elementwise). Then

$$P^{(t)} = 1 - \exp(A^{(t-1)}U),$$

where $A^{(t-1)}U$ is a matrix multiplication. This enables GPU-parallel simulation for large $|\mathcal{U}|, |I|$.

2.6 Interventions and oracle ground-truth ATE^*

To obtain interventional ground truth, we run the generator under do-interventions on a chosen treatment item i : under $\text{do}(T_i=1)$ we force $W[:, i] \leftarrow 1$ at initialization, and under $\text{do}(T_i=0)$ we prevent item i from being consumed throughout the run. Let Y_j indicate whether j is consumed (in the post-propagation log). We compute

$$ATE_{i \rightarrow j}^* = \mathbb{E}[Y_j \mid \text{do}(T_i=1)] - \mathbb{E}[Y_j \mid \text{do}(T_i=0)],$$

by averaging over users (and, optionally, repeating with multiple random seeds to reduce Monte Carlo variance).

Reproducibility. The simulator code (including interventions, timestamping, and oracle ATE^* computation) and the exact configurations used for all reported synthetic results are included in our (anonymized) repository and will be released publicly upon acceptance.

2.7 Relationship between estimated scores and oracle causal strength

We further sanity-check how the estimated scores relate to the simulator’s oracle causal strength on linked pairs. Figure 1 shows that raw $\widehat{ATE}_{i \rightarrow j}$ increases approximately linearly with the oracle effect (*left*), consistent with the fact that unstabilized IPW targets the causal effect on its original probability scale in this controlled setting. By contrast, $\widehat{STE}_{i \rightarrow j}$ (*right*) is an uncertainty-normalized discovery score rather than an effect estimator on the original scale, so it preserves the same monotone trend while down-weighting pairs with higher estimated uncertainty (e.g., limited overlap or small support).

3 LLM Labeling Protocol

Here we provide additional details regarding our LLM-based labeling protocol used to obtain external causality labels for directed causal item–item relations in real-world domains. We note that the LLM is used only for evaluation (never for training, propensity modeling, model selection, or hyperparameter tuning).

3.1 Model and Inference Configuration

We use OpenAI gpt-5.2 via the Responses API¹ with deterministic decoding (temperature=0, top_p=1.0; no sampling-based penalties). All labeling runs were executed in January 2026 with the

¹<https://platform.openai.com/docs/api-reference/responses>

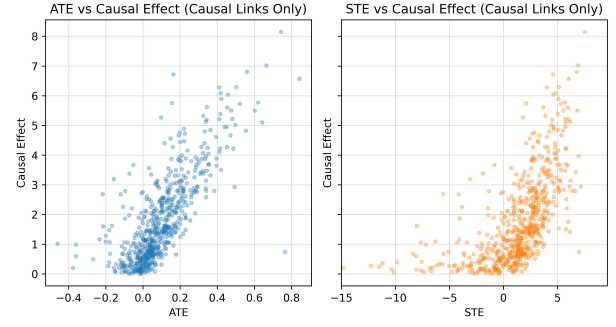


Figure 1: Estimated score versus oracle causal strength on linked pairs in the simulator. Left: $\widehat{ATE}_{i \rightarrow j}$ increases approximately linearly with planted causal strength. Right: $\widehat{STE}_{i \rightarrow j} = \widehat{ATE}_{i \rightarrow j} / \widehat{STD}_{i \rightarrow j}$ preserves the same monotone trend, while down-weighting higher-uncertainty pairs. Each point is a directed item pair.

same fixed inference configuration. Each query requests a structured JSON output and uses a shared ordinal scale `causal_effect` $\in \{0, 1, 2, 3\}$, where 0 indicates no causal mechanism beyond shared audience taste, and 1/2/3 correspond to increasing strength of a direct causal “funnel” (weak/medium/strong; e.g., shared-author spillover vs. strong series linkage vs. explicit sequel/required continuation). The LLM receives only item titles (plus any minimal meta-data already embedded in the dataset title string, e.g., year/edition when present).

Reproducibility. We release the exact prompts, labeling scripts, and configuration files in our anonymized repository (to be made public upon acceptance).

3.2 Candidate Pair Construction

LLM labeling is applied to a fixed set of directed item pairs ($i \rightarrow j$). Accordingly, we randomly select 5,000 pairs from each dataset. Each selected unordered pair $\{i, j\}$ yields two directed queries ($i \rightarrow j$) and ($j \rightarrow i$), which are labeled independently and may receive different scores. All PR/ROC and top- K metrics are computed over this fixed directed candidate set.

3.3 Prompt Template

We use a domain-specific prompt per dataset (books/movies/games). Prompts share the same structure, rubric, and output schema, differing only in domain wording (e.g., *reading* vs. *watching* vs. *playing*) and domain-appropriate mechanisms (e.g., sequels, franchises, expansions, remasters). Below we include the Goodreads prompt; MovieLens and Steam follow the same rubric.

You are labeling directed book pairs (A $\rightarrow$ B) for whether reading A plausibly has a DIRECT causal influence on later reading B, beyond shared reader taste. Causality is directional: A may influence B without the reverse being true.

A “direct causal influence” means there is a plausible mechanism by which exposure to A specifically increases later interest/consumption of B, beyond shared taste correlation. Mechanisms can include (illustrative, not exhaustive): explicit

series continuation (sequels / numbered volumes), prequels commonly read after a main entry, companion/spin-off novels positioned as a next step, short-story collections/novellas that depend on familiarity with the main book, canonical continuations within the same fictional universe, or explicit author-recommended reading order. Any credible direct mechanism beyond shared taste qualifies.

Scoring rubric (0-3):

- 3 (Strong direct influence): Explicit continuation or dependency (clear next step).
- 2 (Medium): Clear funnel or strong linkage, but not an explicit dependency.
- 1 (Weak): Indirect but plausible influence that is more than genre overlap.
- 0 (No direct influence / pure correlation): Only shared taste signals (genre, themes, popularity, same audience) with no specific funnel mechanism.

Rules:

- Do NOT infer causality from genre/theme overlap alone; if that is the only reason, label 0.
- Do NOT infer causality from popularity alone.
- If B was published earlier than A, causal_effect should usually be 0 unless a clear A->B funnel exists (e.g., a prequel commonly read after the main series).
- If uncertain, set causal_effect=0, relation_type="unknown", confidence="low", and say "insufficient evidence".

Return a valid JSON array with exactly one entry per input pair, in the same order. Each entry:

```
{
  "title_A": "<exactly as input>",
  "title_B": "<exactly as input>",
  "causal_effect": 0|1|2|3,
  "relation_type": "sequel"| "prequel"| "spinoff"| "same_series"|
  "same_universe"| "shared_author"| "shared_genre_only"| "other"|
  "unknown",
  "confidence": "low"| "medium"| "high",
  "explanation": "<1-2 sentences>"
}
```

Do not output anything outside the JSON array. Do not include trailing commas.

3.4 LLM Label Distribution

Table 1 reports the distribution of LLM-assigned causal_effect scores over the labeled directed candidate sets. As expected, most pairs receive score 0 (no plausible direct funnel beyond shared taste), while higher scores reflect increasingly explicit mechanisms such as sequels/continuations, franchise ordering, or expansions.

Dataset	0	1	2	3
ML-1M	9862	84	34	20
Steam	9582	286	95	37
Goodreads	9187	520	115	178

Table 1: Distribution of LLM-assigned causal_effect scores over the labeled directed candidate sets (10,000 ordered pairs per dataset, labeled in both directions). Scores range from 0 (no plausible direct influence) to 3 (strong direct influence).

4 Outcome Model (OM) Baseline

This section provides the full specification of the learned outcome-model baseline (OM) used in Sec. 5.3 of the main paper. The role of OM is to serve as a strong shared predictive baseline for downstream item consumption. Unlike ATE/STE, however, OM does not directly estimate a causal estimand; instead, it predicts downstream consumption under a learned parametric model and is evaluated empirically as a ranking baseline. The same model also serves as the outcome-regression component in the doubly robust (DR) baseline reported in the anonymous repository.

4.1 Model definition

For a user u , treatment item i , outcome item j , and treatment indicator $x \in \{0, 1\}$, the model outputs a logit score

$$s(u, i, j, x) = [p_u; q_i]^\top (W + x\Delta) q_j + \alpha_u b_u + \alpha_i b_i + \alpha_j b_j + \alpha_g, \quad (12)$$

where:

- $p_u, q_i, q_j \in \mathbb{R}^d$ are learned embeddings for the user, treatment item, and outcome item;
- $[p_u; q_i] \in \mathbb{R}^{2d}$ denotes concatenation of the user and treatment-item embeddings;
- $W \in \mathbb{R}^{2d \times d}$ is a baseline pathway matrix mapping (u, i) to outcome item j ;
- $\Delta \in \mathbb{R}^{2d \times d}$ is a treatment-specific incremental pathway matrix, active only when $x = 1$;
- $b_u, b_i, b_j \in \mathbb{R}$ are user, treatment-item, and outcome-item bias terms;
- $\alpha_u, \alpha_i, \alpha_j, \alpha_g \in \mathbb{R}$ are learned scalar coefficients.

We use embedding dimension $d = 50$ on MovieLens-1M and $d = 40$ on Goodreads and Steam.

The predicted probability of downstream consumption is

$$m(u, i, j, x) = \sigma(s(u, i, j, x)), \quad (13)$$

where $\sigma(\cdot)$ is the sigmoid function.

Intuitively, W captures the baseline tendency of user u and treatment item i to relate to outcome item j , while Δ captures the additional incremental pathway activated under treatment ($x = 1$). Thus, the model explicitly separates a baseline interaction pathway from a treatment-specific shift.

4.2 Supervised tuple construction

OM is trained on supervised tuples of the form

$$(u, i, j, x, y),$$

constructed from the same post-cutoff user-level treatment/outcome logic used in the main paper. For each ordered pair (i, j) and user u , we define

$$x = T_u^{(i \rightarrow j)} \in \{0, 1\}, \quad y = Y_u^{(j)} \in \{0, 1\},$$

where $T_u^{(i \rightarrow j)}$ and $Y_u^{(j)}$ are exactly the treatment and outcome variables from Eqs. (1)–(2) in the main paper.

Concretely, $x = 1$ iff user u consumed item i after the cutoff and before consuming item j (if j is consumed at all), while $y = 1$ iff user u consumed item j after the cutoff. Thus, OM is trained to answer the supervised question:

given user u , candidate cause item i , candidate effect item j , and treatment status x , what is the probability that j occurs downstream?

This alignment is important: OM is evaluated on the same post-cutoff treatment/outcome construction as ATE/STE, rather than on a separate predictive task.

4.3 Training target and objective

For each supervised tuple (u, i, j, x, y) , the model is trained to predict y from (u, i, j, x) using Bernoulli negative log-likelihood:

$$\mathcal{L}_{\text{OM}} = - \sum_{(u, i, j, x, y) \in \mathcal{D}_{\text{train}}} \left[y \log m(u, i, j, x) + (1-y) \log (1-m(u, i, j, x)) \right]. \quad (14)$$

In practice, the space of all triples (u, i, j) is extremely large, so training uses sampled tuples constructed from the same candidate-pair generation pipeline described in Sec. 5 of the main paper. That is, we first sample candidate ordered pairs (i, j) , then generate user-level supervised examples (u, i, j, x, y) for those pairs from the post-cutoff logs. This yields a shared training/evaluation universe across OM and the other methods.

4.4 Pair scoring

At evaluation time, OM is used as a ranking baseline over ordered item pairs $(i \rightarrow j)$. For each evaluation user u , the model predicts the incremental effect of treatment by comparing the two counterfactual treatment states:

$$\Delta m(u, i, j) = m(u, i, j, 1) - m(u, i, j, 0). \quad (15)$$

We then score the ordered pair (i, j) by aggregating this incremental effect over the evaluation users:

$$\text{OM}(i \rightarrow j) = \frac{1}{N} \sum_{u=1}^N \Delta m(u, i, j), \quad (16)$$

where N is the number of evaluation users considered for that pair.

Thus, OM ranks pairs by their predicted increase in downstream consumption probability under treatment. This makes it a natural learned comparator to ATE/STE, while remaining distinct from the causal-IPW estimators proposed in the main paper. The same scoring construction is also used when OM supplies the outcome-regression term for the DR baseline.

4.5 Interpretation

OM is a shared predictive model, not a semiparametric causal estimator. Its score depends on model specification and training quality, and does not enjoy the same identification interpretation as the IPW-based ATE under the working assumptions of the main paper. We therefore treat OM as a strong learned baseline: if it performs well, this indicates that a parametric predictive model can recover some of the same structure; if it underperforms ATE/STE, this suggests that direct causal-style estimation is better suited for sparse item-item link discovery in organic logs.

5 Additional Results

This section provides supplementary evaluation views beyond the main paper. We first report extended comparisons including all methods from the main paper together with additional variants

omitted there for space. We then provide full Precision-Recall and ROC curves on MovieLens-1M and Steam, full calibration plots for the MF propensity model on all three datasets, and Precision@ K /Recall@ K summaries as fixed- K operating-point views.

5.1 Extended comparisons with additional baselines

The extended tables in this section include all methods from the main paper together with several additional variants omitted there for space. The main-paper methods are: **STE** and **ATE**, our uncertainty-normalized and raw IPW-based causal estimators; **ABLT** and **ABLT/STD**, their propensity-free ablations; **Pearson**, a symmetric correlation baseline; **DoC**, a directional association score based on differences of conditionals; **SASRec**, a sequential prediction baseline; and **OM**, a shared learned outcome model. The additional methods reported only in the auxiliary material are as follows. **Cosine** and **Jaccard** are additional symmetric similarity baselines computed over evaluation users. **DR** is a doubly robust estimator that combines the outcome model and the propensity model, and **DR/STD** is its uncertainty-normalized version. **ATE (stabilized)** and **STE (stabilized)** are the stabilized IPW counterparts of ATE and STE, respectively.

The extended LLM-labeled results table below builds on Table 2 of the main paper with these additional baselines and estimator variants. The broader comparison reinforces the main-paper pattern. On AP, which is our primary metric for rare causal-link discovery, **STE remains the strongest method overall across datasets and thresholds**, while **ATE is typically the runner-up**. This pattern is especially clear at the stricter thresholds, where the labels are more selective and where uncertainty normalization is most useful. The added symmetric similarity baselines (Cosine, Jaccard) remain clearly weaker than STE on AP, even when some of them appear competitive on isolated ROC-AUC blocks. Thus, the auxiliary comparison further supports the main-paper message that broad separability on ROC-AUC is not sufficient for high-precision causal-link discovery. The DR and DR/STD variants underperform both ATE and STE, suggesting that in this sparse pair-discovery setting the variance introduced by residual correction and its uncertainty normalization outweighs their potential bias-reduction benefits. Likewise, the stabilized IPW variants are consistently weaker than their unstabilized counterparts, supporting our choice to use unstabilized ATE/STE in the main paper.

Table 2 extends the sequel-direction benchmark with the same additional baselines and estimator variants. The conclusions again remain unchanged. **STE is strongest overall**, achieving both the highest AP and the highest AUC, while **ATE remains a close runner-up**. Symmetric similarity baselines (Pearson, Cosine, Jaccard) fall exactly at chance, confirming that they cannot recover directionality when both directions of a sequel pair are present. DoC performs substantially worse than chance, revealing its tendency to prefer the reverse direction in this setting. DR-based and stabilized variants also remain weaker than the main IPW-based estimators.

Notably, **ABLT** and **ABLT/STD** are relatively competitive on this benchmark and lie much closer to ATE/STE than they do in the broader LLM-labeled evaluation. This is consistent with sequel direction being driven largely by metadata and temporal order

Method	AP	AUC
STE	0.831	0.821
ATE	0.828	0.815
ABLT/STD	0.820	0.809
ABLT	0.815	0.802
STE (stabilized)	0.822	0.809
ATE (stabilized)	0.744	0.753
DR	0.430	0.394
DR/STD	0.407	0.365
OM	0.592	0.610
SASRec	0.450	0.424
Pearson	0.500	0.500
Cosine	0.500	0.500
Jaccard	0.500	0.500
DoC	0.389	0.336

Table 2: Extended sequel-direction benchmark on Goodreads using metadata-derived labels. Positives correspond to the forward direction within a book series (earlier volume $\rightarrow$ later volume). We report Average Precision (AP) and ROC-AUC, where higher values indicate better recovery of the true sequel direction. The broader comparison reinforces the main-paper pattern: STE is strongest overall, while the benchmark itself emphasizes directionality rather than broad association.

rather than by preference confounding. In such tightly constrained continuation settings, identifying the correct direction is the main challenge, while preference adjustment plays a smaller role. This complements the main-paper interpretation: in the broader LLM-labeled benchmark over diverse item pairs, adjusting for latent user preferences provides a clearer advantage, whereas in sequel data the dominant signal comes from structured continuation relations themselves.

5.2 Extended PR/ROC curves on MovieLens-1M and Steam

Figures 2 and 3 replicate the Goodreads PR/ROC analysis from the main paper on MovieLens-1M and Steam. As in the main paper, we emphasize Precision–Recall and AP because plausibly causal links are rare, and ROC-AUC can appear strong even when precision is low under class imbalance. Consistent with the main results, STE provides the strongest Precision–Recall trade-off, with the clearest gains under stricter thresholds ($t \in \{1, 2\}$), where the LLM-assigned causality labels are most selective. The figures make the same qualitative point as the extended tables: **STE and ATE dominate the Precision–Recall view**, while ROC-AUC is more mixed and therefore less informative for high-confidence discovery.

5.3 Additional calibration plots

Figures 4–6 provide full reliability plots for the MF propensity model on Goodreads, MovieLens-1M, and Steam. Across datasets, predicted propensities track empirical treatment frequencies closely across bins. These plots complement the scalar calibration summaries reported in the main paper and reinforce the role of MF here

as a probabilistic propensity model trained for calibrated absolute probabilities rather than top- K ranking. The close alignment across all three datasets supports the use of MF-derived propensities for IPW-based estimation in this setting.

5.4 Precision@ K /Recall@ K for top- K discovery

Figures 7–9 report Precision@ K and Recall@ K . Precision@ K measures the fraction of positives among the top- K ranked pairs, and Recall@ K measures the fraction of positives recovered within the top- K . Across datasets and thresholds, STE achieves the strongest Precision@ K in the high-confidence regimes, indicating that uncertainty normalization improves the actionable quality of the top-ranked causal hypotheses. In other words, STE is not only strong in threshold-free AP, but also produces the most useful small candidate sets for downstream inspection or intervention.

References

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Method	Goodreads						ML1M						Steam					
	AP (%)			AUC (%)			AP (%)			AUC (%)			AP (%)			AUC (%)		
	>0	>1	>2	>0	>1	>2	>0	>1	>2	>0	>1	>2	>0	>1	>2	>0	>1	>2
STE	46.28	71.18	67.87	73.52	93.41	98.00	18.34	33.17	32.81	70.70	87.60	95.20	30.38	42.63	37.56	72.77	92.18	98.66
ATE	42.96	61.56	60.32	73.50	92.08	97.47	17.96	34.43	31.42	70.63	87.47	92.76	25.52	32.73	34.41	72.40	90.78	98.13
ABLT/STD	39.24	63.16	61.60	68.50	87.09	92.36	4.77	6.97	12.73	63.34	75.74	91.02	16.70	22.12	19.44	54.75	68.18	84.22
ABLT	37.81	60.38	57.88	67.52	85.53	90.97	4.05	5.31	8.26	63.09	74.66	89.65	17.53	23.33	21.27	52.05	62.78	81.23
STE (stabilized)	42.47	64.67	60.86	71.92	92.80	96.75	6.56	8.08	3.83	72.69	86.74	90.22	18.47	19.22	10.54	72.58	91.90	97.23
ATE (stabilized)	31.50	40.58	36.10	70.73	90.38	93.59	3.62	3.60	1.45	70.23	84.39	87.04	9.57	5.67	2.41	70.34	85.41	91.11
DR	22.31	20.46	15.87	69.63	82.93	85.73	11.37	22.51	8.84	67.41	85.12	86.04	12.30	7.67	4.19	68.83	83.01	88.16
DR/STD	14.02	8.71	5.45	65.45	75.86	77.36	5.56	8.80	1.51	66.55	83.90	83.65	4.27	1.55	0.50	53.63	59.92	64.23
OM	29.21	28.56	22.73	70.89	85.25	86.86	5.91	9.73	0.34	57.24	66.30	62.90	21.75	17.40	10.67	72.21	75.22	78.40
SASRec	7.07	2.90	1.74	45.84	51.05	50.23	1.61	0.86	0.67	52.72	62.74	71.72	8.53	2.83	0.88	73.56	74.96	75.78
Pearson	35.66	37.55	32.46	80.30	91.61	93.12	6.69	8.61	4.08	75.19	88.86	87.52	15.33	12.74	6.55	68.28	80.33	91.02
Cosine	22.05	21.34	18.93	72.26	81.30	84.70	2.49	1.90	0.73	67.09	75.06	74.36	15.49	9.98	5.12	76.76	81.67	88.44
Jaccard	32.25	34.01	30.05	75.67	86.63	90.15	5.09	7.09	4.37	68.38	84.25	80.22	15.25	12.96	6.94	68.03	79.22	88.55
DoC	20.56	15.61	10.73	69.04	77.34	79.23	2.50	1.68	0.56	62.01	69.63	58.20	6.88	2.82	0.81	58.96	64.40	63.75

Table 3: Extended results on real-world organic CF logs, including additional symmetric similarity baselines, doubly robust variants, and stabilized IPW variants omitted from the main paper for space. Each method induces a ranking over directed item pairs ($i \rightarrow j$) and is evaluated using deterministic gpt-5.2 ordinal labels `causal_effect` $\in \{0, 1, 2, 3\}$ as an external causality-oriented signal. For each threshold $t \in \{0, 1, 2\}$, positives are defined as `causal_effect` $> t$. We report AP (primary for rare positives) and ROC-AUC; best per block is bold. The broader comparison reinforces the main-paper pattern: STE remains the strongest method overall on AP, while the added variants do not alter the main conclusions.

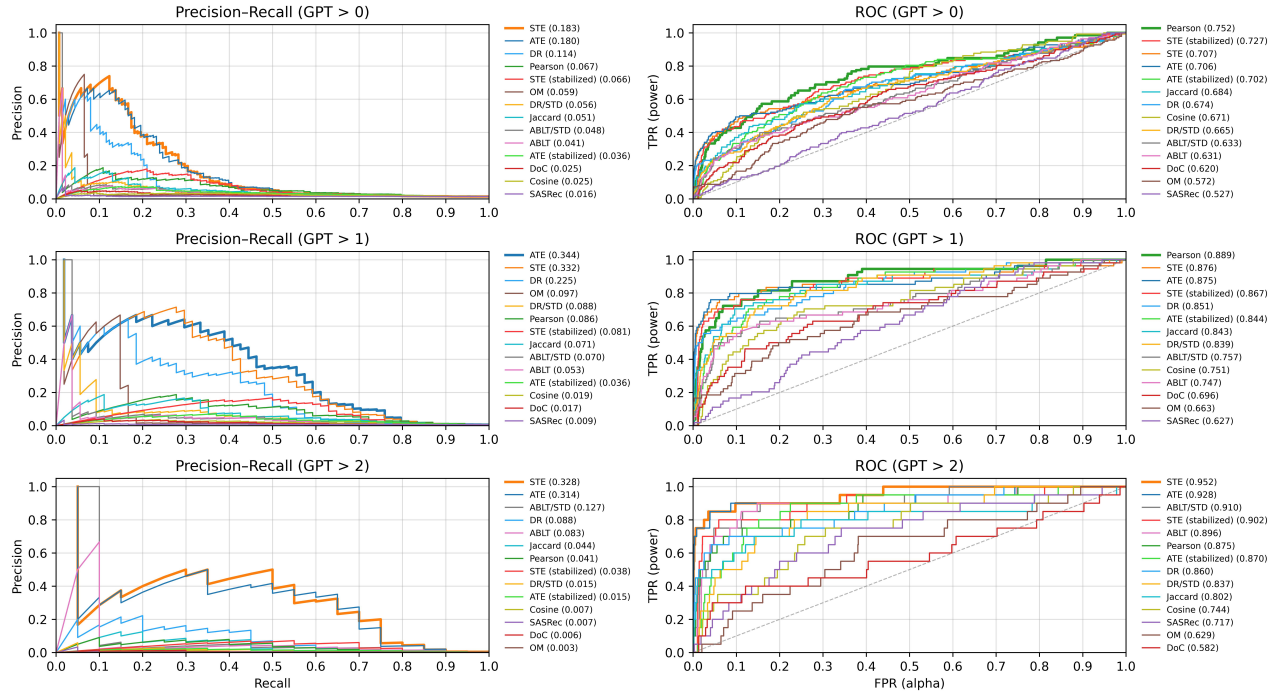


Figure 2: MovieLens-1M PR/ROC. Precision-Recall and ROC curves for ranking directed item pairs. Positives are defined as `causal_effect` $> t$ for $t \in \{0, 1, 2\}$ using LLM-assigned causality labels. Legends report AP and ROC-AUC. As in the main paper, STE provides the strongest Precision-Recall performance, especially at stricter thresholds.

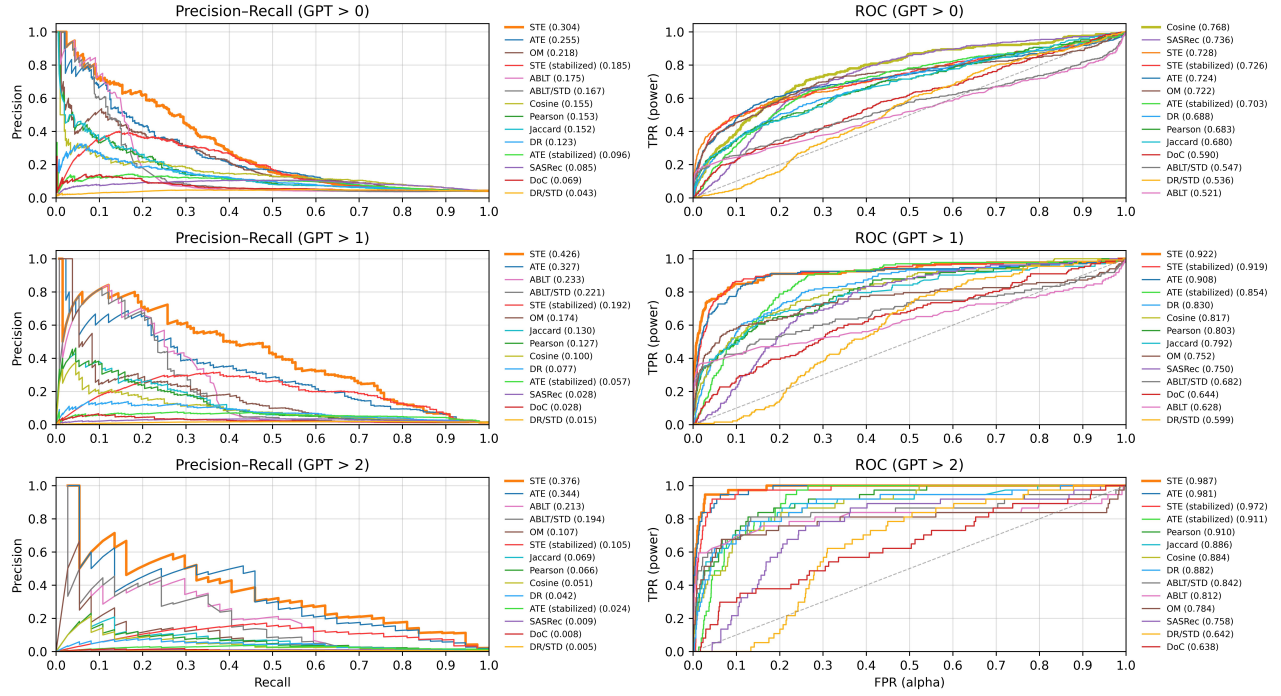


Figure 3: Steam PR/ROC. Precision-Recall and ROC curves for ranking directed item pairs. Positives are defined as $\text{causal_effect} > t$ for $t \in \{0, 1, 2\}$ using LLM-assigned causality labels. Legends report AP and ROC-AUC. As in Goodreads and MovieLens-1M, STE yields the strongest Precision-Recall trade-off.

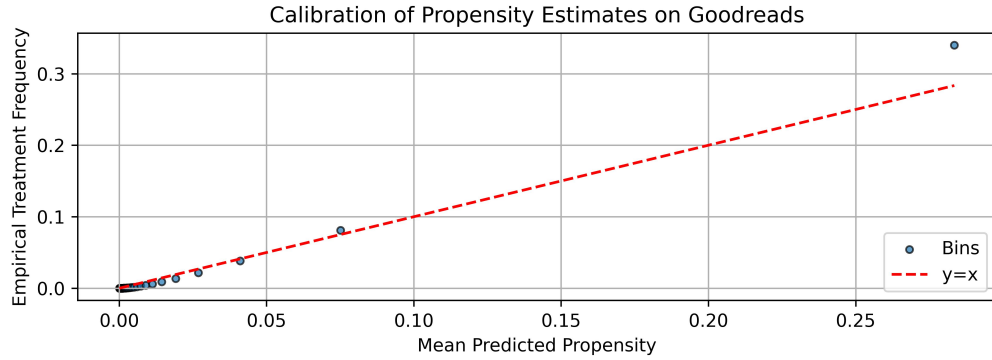


Figure 4: Goodreads calibration. Calibration of the MF propensity model on Goodreads. Each point corresponds to a bin of predicted propensities; the dashed line denotes perfect calibration ($y=x$). MF is used here as a probabilistic propensity model rather than a top- K recommender, since IPW depends on calibrated absolute probabilities.

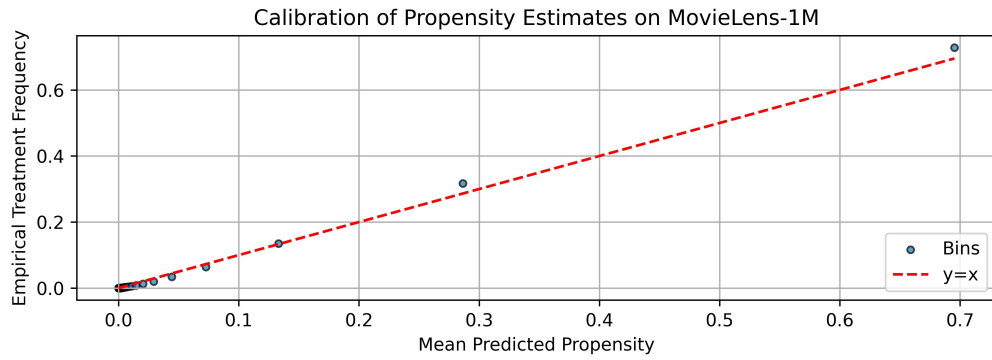


Figure 5: MovieLens-1M calibration. Calibration of the MF propensity model on MovieLens-1M. Each point corresponds to a bin of predicted propensities; the dashed line denotes perfect calibration ($y=x$). MF is used here as a probabilistic propensity model rather than a top- K recommender, since IPW depends on calibrated absolute probabilities.

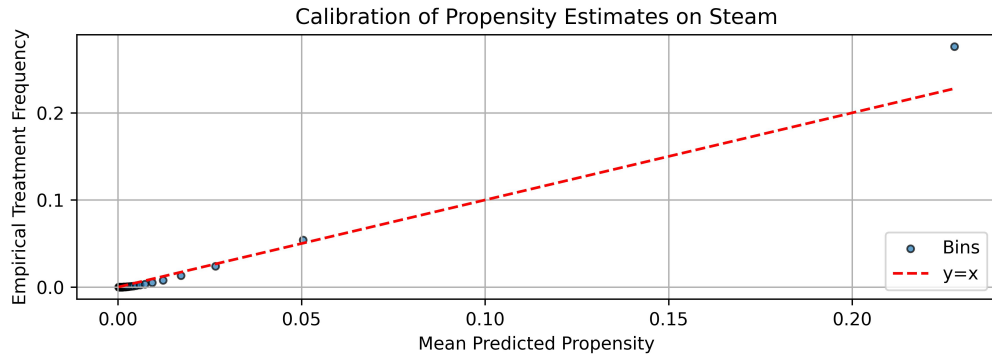


Figure 6: Steam calibration. Calibration of the MF propensity model on Steam. Each point corresponds to a bin of predicted propensities; the dashed line denotes perfect calibration ($y=x$). MF is used here as a probabilistic propensity model rather than a top- K recommender, since IPW depends on calibrated absolute probabilities.

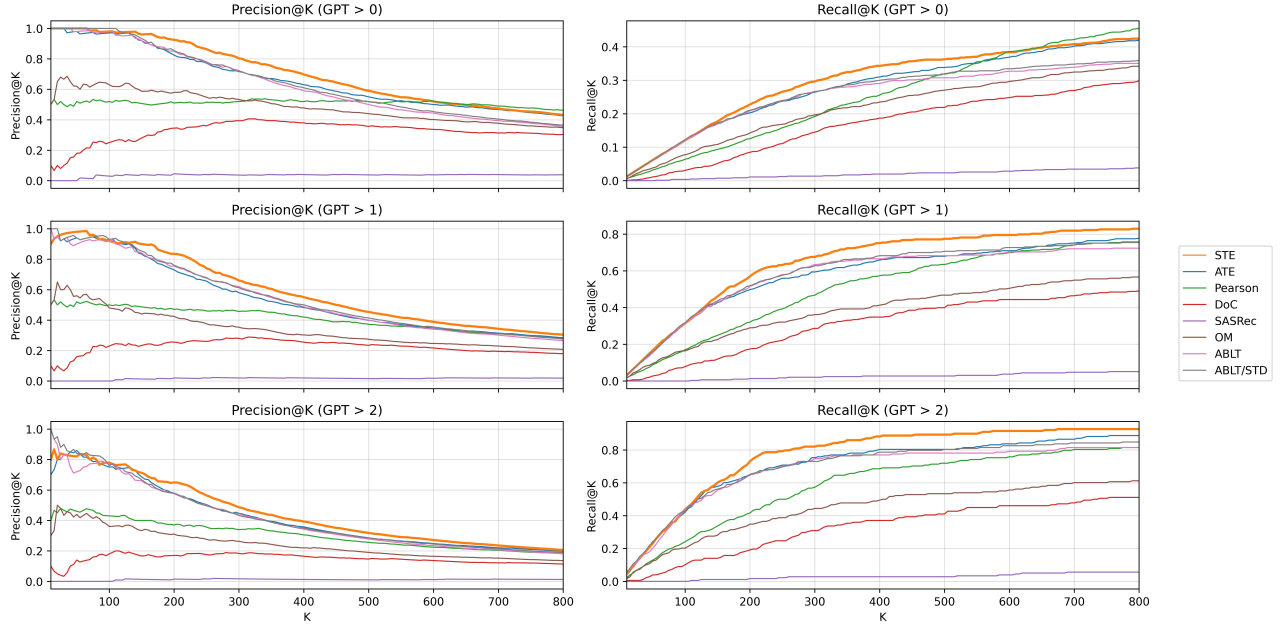


Figure 7: Goodreads Precision@K/Recall@K. Top-K evaluation using LLM-assigned causality labels, with positives defined as $\text{causal_effect} > t$ for $t \in \{0, 1, 2\}$. STE improves the quality of the highest-ranked candidate links, especially under the stricter thresholds.

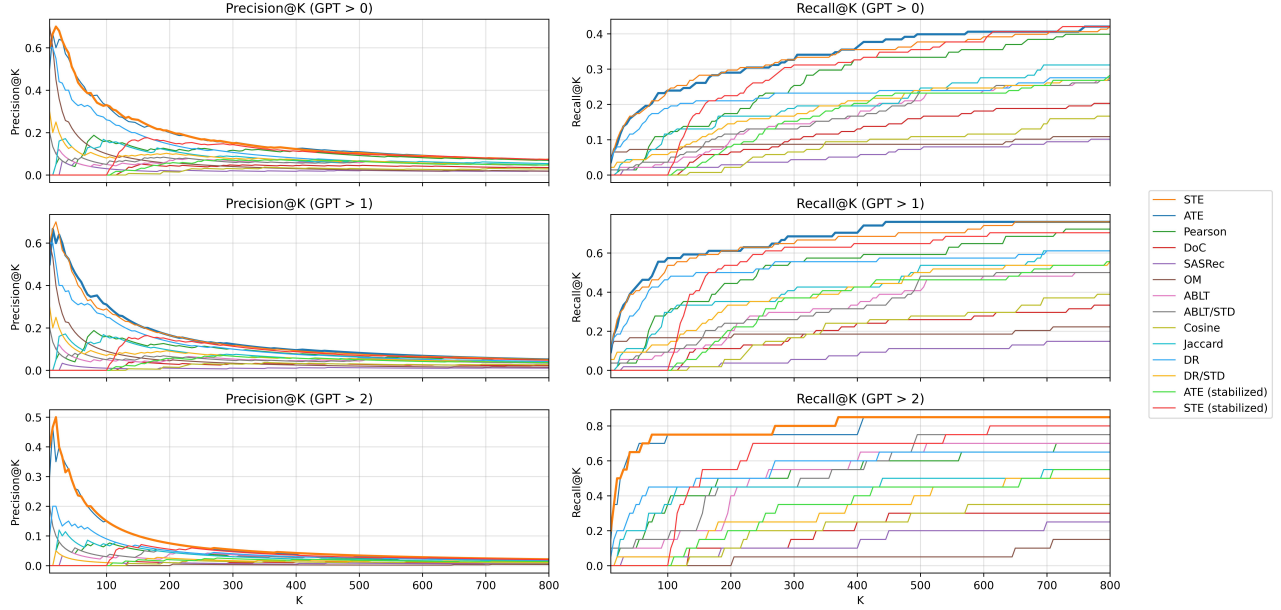


Figure 8: MovieLens-1M Precision@K/Recall@K. Top-K evaluation using LLM-assigned causality labels, with positives defined as $\text{causal_effect} > t$ for $t \in \{0, 1, 2\}$. Consistent with the AP results, STE gives the strongest top-ranked discovery quality.

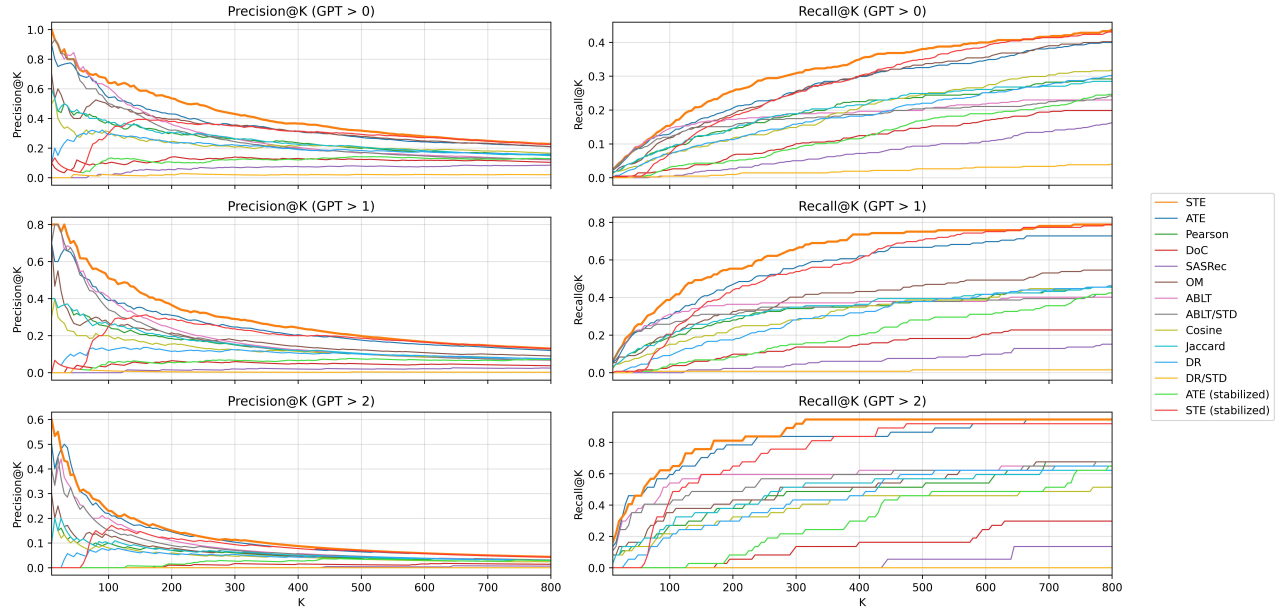


Figure 9: Steam Precision@K/Recall@K. Top-K evaluation using LLM-assigned causality labels, with positives defined as $\text{causal_effect} > t$ for $t \in \{0, 1, 2\}$. STE yields the most actionable top-K candidate sets in the higher-confidence regimes.